One Ensemble Does Not Fit All: Rethinking Query-Free Attacks on Web Vision Services

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Abstract

Vision models exposed through Web APIs and multimodal services are often inaccessible to users beyond their input–output interfaces, complicating the assessment of adversarial transfer across heterogeneous deployments. Transfer-based attacks generate adversarial examples using accessible surrogate models without querying the target during attack construction. To improve transferability, ensemble attacks combine multiple surrogates; however, many established approaches use the same surrogate ensemble for every image, overlooking potential differences in surrogate suitability across inputs. We investigate whether this fixed-composition assumption limits the evaluation of black-box vision systems and introduce DES-Adv, an input-adaptive framework that selects surrogates using representation-space neighborhoods and three complementary signals: local competence, gradient similarity, and gradient sensitivity to input noise. The resulting ensemble can be used with existing attack optimizers. On ImageNet, CIFAR-10, and CIFAR-100, DES-Adv improves transferability against heterogeneous unseen architectures, including improvements of up to 17.64 percentage points in average attack success rate under the evaluated configurations. It also outperforms the 15 fixed four-surrogate combinations from the same six-model pool. We further examine transfer to external image-recognition services, hosted models, and multimodal systems, as well as the effect of adversarial imagery on automated CAPTCHA solvers. Together, these results motivate input-adaptive surrogate composition as a component of robustness evaluation for black-box vision systems used on the Web.

CCS Concepts

• **Computing methodologies** → **Ensemble methods**; *Supervised learning by classification*; *Adversarial learning*.

Keywords

Web Vision Services, Adversarial Transferability, Black-Box Security, Dynamic Ensemble Selection, Surrogate Models

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1 Introduction

Web applications increasingly rely on machine-learning vision components exposed through cloud APIs, hosted inference services, and multimodal interfaces. These services offer convenient access to visual recognition without necessarily revealing the underlying model architectures, weights, or training data. Recent Web security studies have highlighted risks associated with deployed AI services and black-box attack surfaces [12, 19, 33]. This separation creates an important security question: *How reliably can an image crafted using accessible models alter predictions made by a different, inaccessible Web-deployed vision system?* In such settings, relying on repeated target queries may be expensive, rate-limited, or incompatible with the threat model. Prior work has also emphasized the importance of distinguishing realistic attacker capabilities from white-box assumptions [1]. We therefore investigate *query-free transfer attacks*, in which all adversarial optimization is carried out locally on surrogate models and the resulting images are submitted to the target only for evaluation.

Transferability is difficult because deployed systems may use different architectures, optimization procedures, and output interfaces. An adversarial image optimized for one accessible classifier may not transfer to a different target. Ensemble attacks partly address this problem by combining gradients from multiple surrogates; recent analysis highlights the influence of ensemble diversity and model complexity on transferability [31]. Approaches such as SVRE [28], AdaEA [4], and SMER [22] improve how surrogate signals are aggregated or optimized. These representative baselines generally use a fixed set of participating models, even when their weights change. More recent work has explored changing the surrogate collection across optimization iterations or selecting surrogate subsets for universal attacks [13, 29]. Our focus is the distinct setting of *per-image* selection, based on neighborhood-specific evidence, for conventional image-dependent transfer attacks.

We study a complementary design question: *Should surrogate composition itself adapt to the input image?* Rather than search for one globally effective subset of models, we hypothesize that the useful subset can vary with local image characteristics and surrogate gradient behavior. This idea is related to dynamic ensemble selection [5], but our goal is different from improving ordinary ensemble classification. It also differs from prior work that selects training checkpoints [16] or augments vision-transformer surrogates [2], which changes the candidate representations rather than defining our local three-signal selector. We use input-dependent model selection to construct *transferable adversarial perturbations* for unseen targets, without using target feedback to select surrogates.

We propose DES-Adv, an attack-agnostic surrogate-selection framework. For each image, a self-supervised embedding and a nearest-neighbor index identify a local region of competence within a held-out selection set. Candidate surrogates are assessed through local predictive competence, gradient similarity, and gradient sensitivity to small input perturbations. These complementary signals are combined to construct a weighted surrogate subset, which can then be used by an existing ensemble attack. Importantly, these are surrogate-side measurements; the target's weights, gradients, and predictions are not used to select models during attack generation. The independent value of each signal is examined in the ablation study (Section 5.4).

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Our evaluation proceeds from controlled transfer benchmarks to externally deployed systems. We first compare multiple attack algorithms and heterogeneous CNN, Transformer, and MLP target architectures on ImageNet, CIFAR-10, and CIFAR-100. We then evaluate images against cloud vision APIs, hosted inference, and multimodal models to test whether the observed transfer extends beyond the benchmark targets. A separate CAPTCHA study probes automated image solvers with adversarially perturbed visual challenges; this is an application-level robustness test, not a claim to have compromised a production CAPTCHA provider. In the evaluated benchmark settings, dynamic selection improves average attack success rate by up to 17.64 percentage points and exceeds every fixed four-model combination from a six-model pool. These results expose an overlooked variable in black-box robustness assessment: *the input-specific composition of the local surrogate ensemble.*

Contributions. Our contributions are threefold:

- We formulate input-adaptive surrogate composition as a problem in the query-free robustness evaluation of black-box Web vision services and introduce DES-Adv, a modular selection framework based on local competence and gradient behavior.
- We systematically compare dynamic selection with fixed surrogate combinations across architectures, datasets, and four ensemble attack methods, including exhaustive evaluation of all 15 four-model subsets from a six-model pool.
- We study the applicability and limitations of transfer attacks in externally deployed vision APIs, hosted models, multimodal systems, and automated CAPTCHA solvers, highlighting application-specific evaluation metrics and threat-model assumptions.

2 Background and Related Work

Transfer-Based Adversarial Attacks. Adversarial attacks generate small perturbations that cause neural networks to produce incorrect predictions; surveys summarize the broader evolution of attacks on visual tasks [32]. Foundational methods such as FGSM [11], PGD [20], and MI-FGSM [8] optimize adversarial examples using model gradients. In black-box settings, however, direct access to the target is unavailable, motivating transfer-based attacks in which adversarial examples are generated on surrogate models and then transferred to unseen targets. Transferability depends on relationships between surrogate and target decision boundaries, architectures, and gradient directions [6, 15]. Input-diversity and translation-invariant attacks offer complementary ways to improve transfer across models without changing surrogate membership [9, 27]. Consequently, determining which surrogates to use is a distinct design dimension.

Ensemble Transfer Attacks. Using multiple surrogate models can improve transferability by reducing dependence on a single model. EnsAttack [18] combines predictions from several surrogates, while SVRE [28] reduces variance among ensemble gradients. AdaEA [4] introduces adaptive gradient modulation and disparity-reduced filtering to balance model contributions and reduce gradient conflicts. SMER [22] emphasizes surrogate diversity and dynamically reweights ensemble members using reinforcement learning. Complementary work on adversarial example soups combines generated perturbations rather than selecting the participating surrogate models [30]. A recent theoretical study identifies model vulnerability, diversity, and complexity as interacting factors in ensemble transferability [31].

Adaptive Surrogate Construction and Selection. Recent work has begun to relax the assumption of a fixed surrogate collection. Selective Ensemble Attack (SEA) changes the subset of models *across optimization iterations* to exploit a larger pool under a constrained per-step budget [29]; X-Transfer dynamically scales surrogates for universal perturbations targeting CLIP and downstream vision–language systems [13]. ViT-EnsembleAttack augments the candidate Vision Transformers [2], whereas checkpoint-based attacks select training snapshots of a single surrogate [16]. Unlike SEA, which changes surrogate membership during attack optimization, DES-Adv selects an image-specific subset beforehand using local competence, gradient diversity, and noise sensitivity. It complements adaptive aggregation methods by preserving the underlying attack's weighting and optimization procedures.

Security of Web-Hosted AI Systems. Recent studies show the importance of connecting adversarial methods to actual Web deployment assumptions. LENS [19] examines vulnerabilities in self-hosted LLM services, while ICL-Evader [12] studies zero-query black-box evasion of in-context classifiers. MAVLA [33] investigates multimodal adversarial perturbations in vision–language–action systems distributed through an open-Web model ecosystem. Our work targets a distinct problem: selecting image-surrogate ensembles without using feedback from an unseen Web vision target. We evaluate whether this surrogate-side adaptation transfers across prediction APIs and multimodal interfaces; unlike service probing, the attack-generation procedure does not interact with the target.

Dynamic Ensemble Selection. Dynamic Ensemble Selection (DES) addresses a related problem in ensemble learning by selecting classifiers according to their competence in the neighborhood of each test sample [5]. Rather than assuming one static ensemble is optimal globally, DES exploits local variation in classifier behavior. Although competence-based selection is established for prediction ensembles, its use as a criterion for constructing image-dependent, transferable attack ensembles remains less studied. DES-Adv bridges this gap by adapting DES to surrogate-model selection: a representation-space region of competence is combined with local accuracy and gradient-based criteria to dynamically construct an input-specific surrogate ensemble. Unlike prior ensemble attacks that adapt gradient aggregation or weights within a predefined surrogate set, DES-Adv dynamically determines which surrogates participate for each input.

3 DES-Adv: Method

Target System and Threat Model. We consider an image-consuming Web service or hosted model F whose internals are unavailable. The attacker controls the submitted image and can access an independent pool of surrogate models C and a separate labeled selection set D_{DSEL}. Adversarial images are constructed locally without querying F or using target gradients, weights, or output feedback for model selection or perturbation optimization. Target queries are used only in the subsequent evaluation. Unless otherwise noted, our experiments measure untargeted prediction changes; they do not establish downstream account compromise or a successful attack against a production CAPTCHA deployment. The benchmark

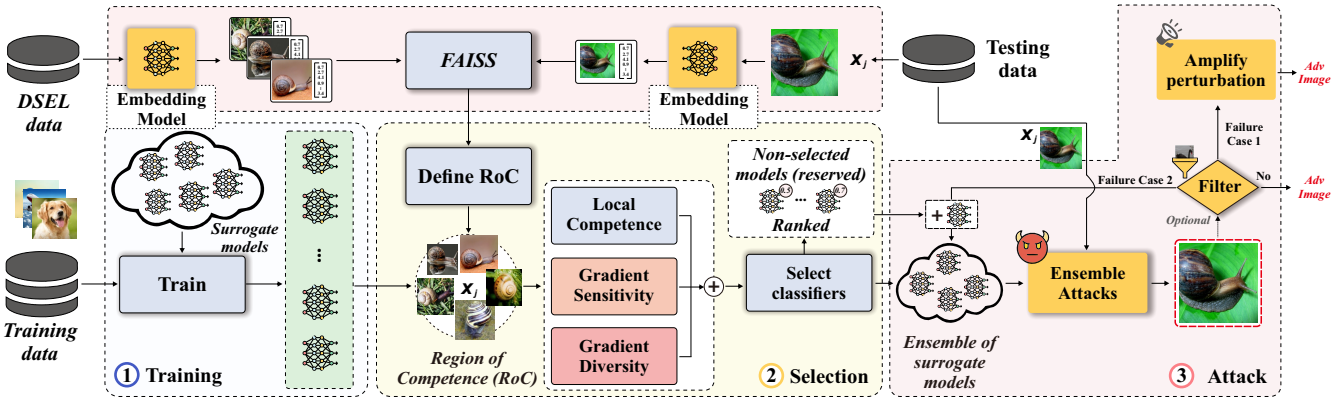


Figure 1: Overview of DES-Adv. A region of competence supplies the three selection signals, which jointly determine the input-specific surrogate subset before the ensemble attack. The failure-case refinement modules shown on the right are *optional*, are disabled in all primary comparisons, and are evaluated separately in the failure-case study. No target-service feedback is used for selection or refinement.

assumes access to ground-truth image labels, an assumption that must be distinguished from fully label-free attacks.

Given a candidate pool of surrogate models, DES-Adv dynamically selects an input-specific subset before applying an ensemble attack. Figure 1 and Algorithm 1 summarize the overall procedure.

Training of Surrogate Models. We assemble a diverse pool of pretrained and, where noted, independently trained surrogate models to encourage variation in architecture, capacity, and decision boundaries, which can influence transferability [4, 22]. A separate selection set (DSEL) is held out for defining regions of competence and estimating surrogate competence during attack generation.

Region of Competence. We first extract embeddings for all samples in the validation set D_{DSEL} using the feature extractor $f(\cdot)$ and build a FAISS index [10] for efficient nearest-neighbor search. For a target input t with embedding $e_t = f(t)$, we obtain the k most similar validation samples:

$$\psi_t = \text{FAISS_Query}(e_t, k), \tag{1}$$

which forms the local RoC around t.

Local Surrogate Competence. Given the RoC ψ_t, we evaluate each candidate surrogate $c_i \in C$ using three complementary signals before making the final subset selection. The local competence of each surrogate is computed as its classification accuracy over the retrieved neighbors:

$$comp_i(t) = \frac{1}{|\psi_t|} \sum_{(x_j, y_j) \in \psi_t} \mathbb{I}\left[\arg\max_c c_i(x_j)_c = y_j\right], \tag{2}$$

where $\mathbb{I}[\cdot]$ is the indicator function and y_j denotes the ground-truth label of neighbor x_j. All retrieved neighbors contribute equally, without similarity-based weighting. Local competence serves as one of the three selection criteria rather than an independent selection rule.

Gradient Diversity. To characterize the diversity of surrogate decision landscapes, we examine the alignment of input gradients across candidate surrogate models. For each surrogate $c_i \in C$, we compute its input gradient using cross-entropy loss:

$$g_i(t) = \nabla_t \mathcal{L}_{\text{CE}}(c_i(t), \hat{y}_i), \qquad \hat{y}_i = \arg\max_c c_i(t)_c, \tag{3}$$

where $\hat{y}_i$ denotes the surrogate's predicted class.

We then measure the average cosine similarity between the gradient of surrogate c_i and those of all candidate surrogates:

$$G_i(t) = 1 - \frac{1}{|C|} \sum_{j=1}^{|C|} \frac{g_i(t)^\top g_j(t)}{\|g_i(t)\|_2 \|g_j(t)\|_2}. \tag{4}$$

A higher value indicates lower average gradient alignment with the candidate pool, reflecting greater inter-surrogate gradient diversity. This criterion provides a complementary signal to local predictive competence by characterizing differences in the surrogates' input-loss gradients.

Gradient Instability (Noise Sensitivity). While gradient diversity characterizes the alignment of gradients across different surrogate models, it does not capture how individual surrogate gradients respond to small input perturbations. To characterize this behavior, we introduce a gradient instability criterion.

For each candidate surrogate $c_i \in C$, we perturb the input image with Gaussian noise:

$$\tilde{t} = \text{Clip}_{[0,1]}(t + \eta), \qquad \eta \sim \mathcal{N}(0, \sigma^2 I). \tag{5}$$

We then compute the input gradient at the perturbed image using the same reference label $\hat{y}_i$ as the original input:

$$g_i'(t) = \nabla_{\tilde{t}} \mathcal{L}_{\text{CE}}(c_i(\tilde{t}), \hat{y}_i). \tag{6}$$

Gradient instability is measured as

$$stab_i(t) = 1 - \text{sim}(g_i(t), g_i'(t)), \tag{7}$$

where $\text{sim}(\cdot)$ denotes cosine similarity. A higher value indicates greater sensitivity of the surrogate's input gradient to small perturbations. Unlike inter-surrogate gradient diversity, this criterion captures local variation within an individual surrogate's loss landscape. We treat gradient instability as an empirical selection signal rather than a theoretically guaranteed indicator of adversarial transferability.

Joint Surrogate Scoring and Selection. For each candidate c_i, we first map the three selection signals to $[0, 1]$:

$$\mathbf{q}_i(t) = \left(comp_i(t),\ \frac{G_i(t)}{2},\ \frac{stab_i(t)}{2} \right). \tag{8}$$

We employ an input-adaptive coefficient assignment to combine the three selection signals. Starting from reference coefficients $\mathbf{a}^0 = (\alpha_0, \beta_0, \gamma_0)$, we adjust their relative contributions according to the distribution of signal scores across the candidate surrogate pool. Let $d_j(t) = \max_i q_{ij}(t) - \min_i q_{ij}(t)$ denote the range of signal j across the candidate pool. Define $r_j(t) = d_j(t)/\sum_h d_h(t)$ when the total range is positive and $r_j(t) = 0$ otherwise. We then calculate

$$a_j(t) = \frac{a_j^0 \left(1 + \lambda r_j(t)\right)}{\sum_h a_h^0 \left(1 + \lambda r_h(t)\right)}, \qquad 0 \le \lambda \le 1. \tag{9}$$

The candidate score is $s_i(t) = \mathbf{a}(t)^\top \mathbf{q}_i(t)$; the m highest-scoring candidates form

$$E_t^* = \mathrm{TopM}_{c_i \in C}\ s_i(t). \tag{10}$$

Importantly, the proposed scoring function is used exclusively to rank and select surrogate models. The selected subset E_t^* is passed directly to the underlying ensemble attack, without modifying its original gradient aggregation or model-weighting mechanism. Thus, any adaptive weighting performed during adversarial optimization is determined solely by the chosen attack algorithm, not by DES-Adv.

Optional Failure-Case Refinement. We additionally investigate two optional, surrogate-side refinement heuristics. These refinements are disabled in all primary ImageNet, CIFAR, and external-service comparisons and are evaluated only in the dedicated failure-case study (Appendix D). They use observable properties of the generated perturbation to flag potential transfer-risk cases, not feedback from the inaccessible target. The optional behavior is controlled by $r_{\text{refine}} \in \{0, 1\}$, with $r_{\text{refine}} = 0$ for the main experiments. First, we identify an *underutilized-budget case* when the generated perturbation underutilizes the allowed attack budget defined by $\|\delta\|_p \le \epsilon$, where $\delta = x_{\text{adv}} - x$ denotes the adversarial perturbation, $\|\cdot\|_p$ is the ℓ_p norm, and ϵ is the maximum perturbation magnitude controlling the attack strength. Specifically, we detect such cases when $\|\delta\|_p/\epsilon < \tau_{\text{noise}}$, where τ_{noise} is a predefined threshold controlling the minimum relative perturbation strength, used to detect underutilized perturbation budgets, indicating insufficient perturbation strength. This typically corresponds to cases where the injected noise magnitude is significantly lower (approximately three times smaller) than that observed in successful attacks. For low-perturbation cases, we amplify the existing perturbation by the largest feasible factor $\kappa \in \{1, 2, 3\}$ satisfying $\kappa\|\delta\|_\infty \le \epsilon$. Second, we detect a *clean-background failure* when the perturbation is predominantly concentrated in low-texture regions. Concretely, we compute the mean gradient magnitude over perturbed pixels,

$$T = \frac{1}{|\Omega|} \sum_{u \in \Omega} G(x_u),$$

and trigger this condition when $T < \tau_{\text{texture}}$, where τ_{texture} is a predefined threshold estimated from the lower percentile of surrogate-side texture statistics. In such cases, we augment the ensemble by introducing a reserved surrogate: $E_t^* \leftarrow E_t^* \cup c_{\text{new}}$, and regenerate the adversarial example using Eq. (11). We analyze these heuristics separately in Appendix D; because noise amplification can change the perturbation budget, its results must not be conflated with equal-budget selection improvements.

Algorithm 1 DES-Adv: Three-Signal Surrogate Selection with Optional Refinement

Input: C: surrogate pool, D_{DSEL}: selection set, t: input, f: embedding function, k: neighbors, m: selected surrogates, $\mathbf{a}^0$: reference signal coefficients, λ: coefficient adaptation strength, r_{refine}: optional refinement (default 0).
Output: Adversarial image x_{adv}.

$\psi_t \leftarrow \text{FAISS_Query}(\text{BuildFAISS}(f(D_{\text{DSEL}})), f(t), k)$
$\tilde{t} \leftarrow \text{Clip}_{[0,1]}(t + \eta), \quad \eta \sim \mathcal{N}(0, \sigma^2)$
// Compute local competence and gradients
for each $c_i \in C$ **do**
  $comp_i \leftarrow \text{ComputeCompetence}(c_i, \psi_t)$
  $\hat{y}_i \leftarrow \arg\max_c c_i(t)_c$
  $g_i \leftarrow \nabla_t \mathcal{L}_{\text{CE}}(c_i(t), \hat{y}_i)$
  $g_i' \leftarrow \nabla_{\tilde{t}} \mathcal{L}_{\text{CE}}(c_i(\tilde{t}), \hat{y}_i)$
  $stab_i \leftarrow 1 - \text{sim}(g_i, g_i')$
end for
// Compute inter-surrogate gradient diversity
for each $c_i \in C$ **do**
  $G_i \leftarrow 1 - \frac{1}{|C|} \sum_{c_j \in C} \text{sim}(g_i, g_j)$
  $\mathbf{q}_i \leftarrow (comp_i, G_i/2, stab_i/2)$
end for
// Adaptive signal combination
$d_j \leftarrow \max_i q_{ij} - \min_i q_{ij}$ for $j \in \{C, G, S\}$
$\mathbf{r} \leftarrow \mathbf{d}/\sum_j d_j$ if $\sum_j d_j > 0$, else $\mathbf{r} \leftarrow \mathbf{0}$
$\mathbf{a}(t) \leftarrow \text{Normalize}(\mathbf{a}^0 \odot (\mathbf{1} + \lambda \mathbf{r}))$
$s_i \leftarrow \mathbf{a}(t)^\top \mathbf{q}_i$ for each $c_i \in C$
$E_t^* \leftarrow \text{TopM}_{c_i \in C}(s_i, m)$
// Original ensemble attack
$x_{\text{adv}} \leftarrow \text{EnsembleAttack}(E_t^*)$
if $r_{\text{refine}} = 1$ **then**
  // Optional; excluded from primary results
  if $\text{LowNoiseFailure}(x_{\text{adv}})$ **then**
    $x_{\text{adv}} \leftarrow \text{AmplifyNoise}(x_{\text{adv}})$
  end if
  if $\text{CleanBackgroundFailure}(x_{\text{adv}})$ **then**
    $c_{\text{new}} \leftarrow \text{NextReservedModel}(C \setminus E_t^*)$
    $E_t^* \leftarrow E_t^* \cup \{c_{\text{new}}\}$
    $x_{\text{adv}} \leftarrow \text{EnsembleAttack}(E_t^*)$
  end if
end if
return x_{adv}

$$x^{\text{adv}} = \text{EnsembleAttack}(E_t^*), \tag{11}$$

4 DES-Adv: Experimental Settings

In this section, we describe the experimental setup used to evaluate adversarial attack transferability. We provide details on the datasets, models, attack configurations, evaluation metrics, and hardware environment to ensure reproducibility. All primary quantitative comparisons use the core three-signal selector ($r_{\text{refine}} = 0$); the two optional failure-case refinements are studied only in Appendix D.

4.1 Evaluation Metrics

Attack Success Rate (ASR). ASR measures the ratio of adversarial examples that successfully fool the model, *i.e.,* the percentage of adversarial inputs that cause a misclassification. It is defined as:

$$\text{ASR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[F(x_i^{\text{adv}}) \neq y_i\right],$$

For external Web services, we report conditional Top-k ASR: the proportion of benign-correct inputs for which the ground-truth class is absent from the adversarial Top-k predictions. Benign correctness is determined using a consistent label-matching procedure,

Table 1: Details of Target Models.

Model	BiT-101	ResNet-152	WRN101-2	RegNet-160	RegNet-320	VGG-19	ViT-B	DeiT-B	Swin-B	MLP-Mixer-B	ConvMixer
Type	CNN	CNN	CNN	CNN	CNN	CNN	ViT	ViT	ViT	MLP	CNN
Dataset	CIFAR-10/100	All	ImageNet	CIFAR-10/100	ImageNet	ImageNet	All	All	All	ImageNet	All

Table 2: Details of Surrogate Models across ImageNet, CIFAR-10, and CIFAR-100 (accuracy on 1000 sampled test images).

Model	**ImageNet**	**CIFAR-10**	**CIFAR-100**	**Type**
ResNet-18	76.7	93.40	78.05	CNN
Inception-V3	74.4	92.76	78.81	CNN
EfficientNet-B0	91.8	93.37	83.56	CNN
XCiT-Tiny	96.3	–	–	ViT
DeiT-Tiny	80.6	95.73	75.26	ViT
ViT-Tiny	86.3	96.66	80.53	ViT
GCViT-Tiny	–	93.05	87.68	ViT

and all compared attacks share the same eligible inputs. Consequently, the evaluation denominator depends on the service and k, rather than the total test-set size. We provide matching rules and effective sample counts in the appendix.

Noise Rate. We measure perceptual distortion as $1 - \text{SSIM}$ [25], where higher values indicate greater distortion.

$$\text{SSIM}(x, x') = \frac{(2\mu_x\mu_{x'} + C_1)(2\sigma_{xx'} + C_2)}{(\mu_x^2 + \mu_{x'}^2 + C_1)(\sigma_x^2 + \sigma_{x'}^2 + C_2)}$$

Pixel Difference (PD). We measure the mean absolute pixel difference between clean and adversarial images after clipping to $[0, 1]$: constraints as follows.

$$\text{PD} = \frac{1}{NHWC} \sum_{i,h,w,c} \left| \text{CLAMP}(x_i^{\text{adv}})_{h,w,c} - x_{i,h,w,c} \right|.$$

Attack Time. We report the total time (minutes) to generate 1,000 adversarial images.

4.2 Datasets and Models

Datasets. We evaluate on ImageNet [7] and CIFAR-10/100 [14], following prior transfer-attack studies [4, 22]. For each dataset, we sample 1,000 images correctly classified by all targets, enabling consistent ASR comparisons. Thus, reported ASR is conditional on joint benign correctness. Target predictions are used only for test-set construction, never for surrogate selection or attack optimization.

Surrogate and Target Models. We use six diverse surrogates per dataset (Table 2). ImageNet models are obtained from `timm` [26], while CIFAR-10/100 models are sourced from AdaEA [4], except starred models trained by us. Reported accuracies are measured on the selected 1,000-image subsets. We evaluate nine ImageNet and seven CIFAR target models spanning CNNs, Transformers, MLP-Mixer, and ConvMixer (Table 1). Unlike MLP-Mixer, ConvMixer is convolutional [24].

Attack Configuration. We evaluate four ensemble attack methods: EnsAttack, SVRE, AdaEA, and SMER, using MI-FGSM and PGD optimization. Unless otherwise specified, we use an ℓ_∞ budget of $\epsilon = 8/255$, step size $\alpha = 2/255$, and 10 iterations, with inputs clipped to $[0, 1]$. MI-FGSM uses momentum $\mu = 1.0$. AdaEA uses $\tau = -0.3$, while AdaEA and SMER use $\beta = 10$.

DES-Adv Configuration. We use DINO ViT-Base/16 [3] embeddings with L2 normalization and FAISS cosine retrieval ($k = 7$). DSEL contains 30,000 ImageNet and 14,000 CIFAR-10/100 samples. Surrogates are ranked using the three signals and adaptive coefficients in Section 3. Main results report mean ± standard deviation over five runs, while single-run ablations are interpreted descriptively. Experiments use three RTX 2080 Ti GPUs, an Intel Xeon E5-2620 v3 CPU, CUDA 13.0, and Python 3.10.19.

5 DES-Adv: Results and Discussions

5.1 Transferability of Attacks

Table 3 reports the transferability of individual surrogates to unseen targets. Under the evaluated settings, MI-FGSM outperforms PGD, with DeiT-Tiny achieving the highest average ASR (47.51%). Transformer-based surrogates generally transfer better to Transformer targets, while CNN-based surrogates perform better on CNN targets. DeiT-Tiny and EfficientNet-B0 show strong transferability across architectures, consistent with their competitive classification accuracy (Table 2).

5.2 Ensemble Attack Performance

We evaluate three fixed four-surrogate configurations drawn from a six-model pool (Table 8, first three rows) to assess the impact of surrogate composition on attack transferability.

Figure 2 shows that SMER achieves the highest average ASR under MI-FGSM (81.77%) and PGD (77.76%) with Set 3, albeit at higher computational cost and slightly greater noise. Sets 2 and 3 generally outperform Set 1, highlighting the importance of surrogate composition. Detailed per-target results are reported in Appendix Table 10.

DES-Adv for Ensemble Attacks. Figure 2 also evaluates DES-Adv, which dynamically selects four surrogates from a six-model pool for each input before applying the underlying ensemble attack. Across the evaluated ensemble attack methods, DES-Adv consistently improves average ASR over the three predefined fixed surrogate configurations. For instance, under MI-FGSM with Ens. Attack, DES-Adv achieves an average ASR of 79.66%, corresponding to an improvement of 9.16 percentage points over the strongest predefined fixed ensemble. Similarly, under PGD with SVRE, DES-Adv achieves 71.61% average ASR, improving upon the strongest predefined configuration by 17.64 percentage points. These results demonstrate the potential of input-adaptive surrogate selection to improve attack transferability across different ensemble optimization strategies, with only minor changes in noise rate and pixel difference.

To further distinguish input-adaptive selection from the choice of a particular fixed ensemble, we additionally evaluate all 15 possible fixed four-surrogate combinations from the same six-model pool. The exhaustive comparison is presented in Figure 4.

Table 3: Transferability between surrogate and target models on ImageNet.

Attack	Sur. / Tar.	Res-152	WRN101-2	RegNet-320	VGG-19	ViT-B	DeiT-B	Swin-B	MLP-Mixer-B	ConvMixer	Avg ASR	Noise Rate	Pixel-diff
MI-FGSM	ResNet-18	38.10	55.10	27.30	77.20	10.80	16.10	13.80	31.20	45.80	35.04	0.194 ± 0.082	0.025 ± 0.001
	Inception-V3	26.30	42.00	20.20	57.80	9.80	15.20	10.30	26.80	34.70	27.01	0.183 ± 0.077	0.025 ± 0.001
	DeiT-Tiny	25.10	44.40	22.70	65.10	51.00	77.20	26.40	72.80	42.90	**47.51**	0.190 ± 0.081	0.026 ± 0.001
	ViT-Tiny	20.90	39.90	21.30	60.30	49.80	62.00	25.70	59.90	38.00	41.98	0.191 ± 0.081	0.026 ± 0.001
	EfficientNet-B0	44.40	65.00	43.80	80.60	22.10	33.70	31.30	47.90	58.70	47.50	0.197 ± 0.083	0.026 ± 0.001
	XCiT-Tiny	30.50	41.10	41.30	70.10	17.30	22.90	26.20	34.60	46.00	36.67	0.200 ± 0.083	0.026 ± 0.001
PGD	ResNet-18	21.70	36.80	14.80	60.50	4.70	7.30	7.40	18.10	27.10	22.04	0.128 ± 0.060	0.018 ± 0.001
	Inception-V3	12.60	26.50	9.70	40.00	4.70	6.80	5.20	15.90	20.00	15.71	0.123 ± 0.059	0.017 ± 0.001
	DeiT-Tiny	10.40	26.40	10.40	46.10	28.90	54.00	12.90	48.30	23.20	**28.96**	0.126 ± 0.059	0.018 ± 0.001
	ViT-Tiny	9.70	25.80	9.00	42.60	27.90	41.70	13.40	39.30	22.20	25.73	0.127 ± 0.059	0.018 ± 0.001
	EfficientNet-B0	23.00	41.00	24.10	64.70	9.20	14.90	15.20	28.70	36.70	28.61	0.128 ± 0.060	0.018 ± 0.001
	XCiT-Tiny	16.20	24.60	21.60	51.60	7.00	9.30	12.80	21.30	25.50	21.10	0.127 ± 0.060	0.017 ± 0.001

Ensemble Attacks on CIFAR-10/100. We evaluate three surrogate sets on CIFAR-10/100 (Table 9). Figure 3(a) presents the CIFAR-10 results across four ensemble attacks. Among the baselines, EnsAttack achieves the highest average ASR of 83.51% with Set 1. On CIFAR-10, DES-Adv improves average ASR by 7.29, 5.89, 6.23, and 12.33 percentage points for EnsAttack, SVRE, AdaEA, and SMER, respectively. On CIFAR-100, SMER with Set 3 achieves the highest fixed-ensemble ASR (85.66%). As shown in Figure 3(b), DES-Adv improves all four ensemble attacks by 4.65–7.17 percentage points, although gains are smaller than on CIFAR-10.

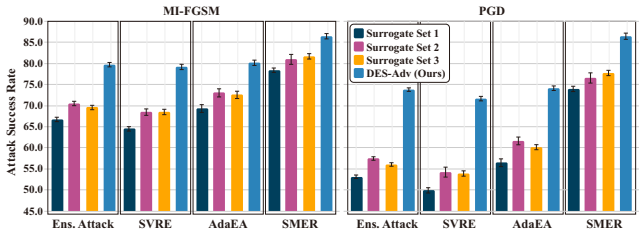


Figure 2: Average ASR (%) on ImageNet across nine unseen targets. Three fixed surrogate sets are compared with DES-Adv under MI-FGSM (left) and PGD (right) with four ensemble attack methods.

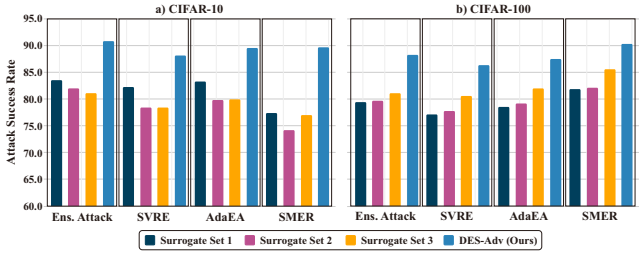


Figure 3: Average ASR (%) under MI-FGSM across seven unseen targets on (a) CIFAR-10 and (b) CIFAR-100, comparing three fixed surrogate sets with DES-Adv across four ensemble attack methods.

Exhaustive Fixed-Subset Comparison. We compare DES-Adv against all 15 fixed four-surrogate subsets from a six-model pool across three datasets and four ensemble attacks (Figure 4). DES-Adv achieves higher average ASR than every fixed subset, supporting input-adaptive selection. Moreover, finding the best fixed subset requires evaluating $\binom{|C|}{m}$ combinations, whereas DES-Adv ranks surrogates directly without exhaustive subset search. Subset compositions are listed in Appendix Tables 8 and 9.

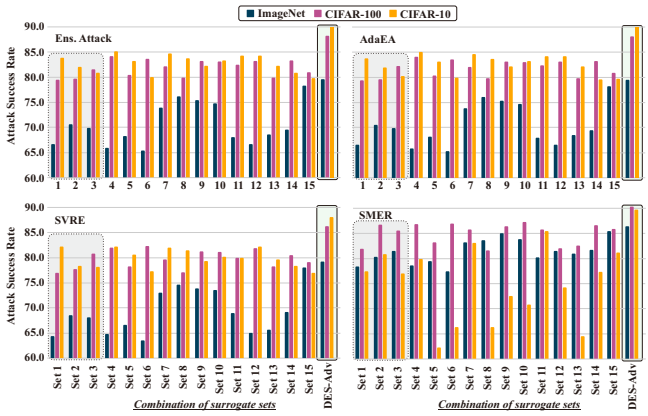


Figure 4: Average transfer ASR (%) for 15 fixed four-surrogate subsets (Sets 1–15) and DES-Adv on ImageNet, CIFAR-10, and CIFAR-100 using MI-FGSM with Ens. Attack, SVRE, AdaEA, and SMER. Set compositions are listed in Tables 8 and 9.

Optional Transfer-Failure Refinements. Beyond the primary three-signal selection results, we investigate two surrogate-observable transfer-risk patterns in dedicated experiments on additional ImageNet samples. For underutilized perturbations, amplifying the generated noise increases transfer success, but also raises the perturbation magnitude, so this is *not* evidence of an equal-budget improvement over the main methods. For visually simple backgrounds, a reserved surrogate is added to the selected ensemble, but the enlarged ensemble changes computational cost and pool size. These optional refinements are not activated in any primary benchmark or Web-service result. The full case definitions, example images, and separate experiments are reported in Appendix D.

The preceding experiments establish transferability across controlled classification benchmarks and unseen target architectures. We next examine whether the gains extend to externally deployed vision services and application-level image-processing pipelines.

5.3 Real-World Transferability Evaluation

Threat Model and Evaluation Protocol. We evaluate query-free transfer against Web vision APIs, hosted classifiers, multimodal systems, and custom CAPTCHA solvers. SMER and SMER+DES-Adv

Table 4: Comparison of SMER and SMER+DES-Adv on real-world image recognition services and hosted models ($N = 1000$ for all services). ASR is computed over benign-correct samples.

Service / Model	Top-1					Top-5					Top-10				
	Ben.	SMER Adv.	SMER ASR	DES-Adv Adv.	DES-Adv ASR	Ben.	SMER Adv.	SMER ASR	DES-Adv Adv.	DES-Adv ASR	Ben.	SMER Adv.	SMER ASR	DES-Adv Adv.	DES-Adv ASR
AWS Rekognition	8.90	2.70	82.02	2.20	**89.89**	47.60	18.80	62.61	13.40	**73.11**	63.90	29.30	55.56	21.70	**66.98**
Google Cloud Vision	11.70	7.70	47.01	5.60	**57.26**	39.50	20.20	51.14	15.90	**62.28**	55.40	30.20	46.93	23.50	**59.57**
HF ConvNeXt-B (Meta)	98.40	31.50	67.99	24.10	**75.51**	100.00	64.60	38.40	55.90	**44.10**	100.00	61.60	38.40	55.90	**44.10**
Imagga	50.90	8.50	85.27	6.40	**88.02**	65.20	15.80	76.84	13.20	**80.67**	70.80	21.30	70.62	18.70	**74.72**

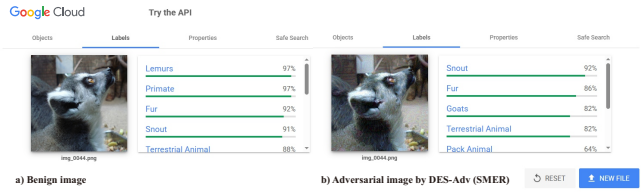


Figure 5: Illustrative transfer attack against Google Cloud Vision. The panels compare benign and adversarial inputs and their returned labels; the adversarial image is generated using SMER+DES-Adv.

use the same 1,000 paired ImageNet images, with optional refinements disabled. Target responses are collected only for evaluation. Since Web APIs return free-form labels, we measure Top-k agreement using normalized text matching; historical cloud-API results use permissive shared-word or substring matching. Conditional ASR is computed over successfully matched, benign-correct pairs. Matching rules and request settings are provided in the evaluation code.

Cloud APIs and Hosted Vision Classification. We evaluate AWS Rekognition (`DetectLabels`, MaxLabels=10, MinConfidence=0), Google Cloud Vision (`LABEL_DETECTION`, 10 labels), Imagga (`/v2/tags`, 10 English tags), and Meta's `facebook/convnext-base-224` via Hugging Face `InferenceClient`. All services are queried remotely. As shown in Table 4, DES-Adv improves SMER's Top-1 ASR on AWS from 82.02% to 89.89%, Top-5 ASR on Google from 51.14% to 62.28%, and Top-1 ASR on Imagga from 85.27% to 88.02%. Coarse API labels (e.g., "animal" or "bird") limit ImageNet label agreement and conditional ASR sample sizes. Figure 5 illustrates benign and adversarial responses.

Multimodal Model APIs. We evaluate `gpt-5.6`, `claude-sonnet-4-6`, and `gemini-3.6-flash` via their respective APIs. Each model is prompted to "identify the main object" and return its five most likely categories in ranked order as JSON, without explanation. Gemini uses temperature zero. As shown in Table 5, DES-Adv improves Top-1 ASR from 60.84% to 75.76% for GPT, 48.30% to 57.62% for Claude, and 26.70% to 40.24% for Gemini.

Custom Adversarial CAPTCHA. We evaluate two Gradio-based challenges: nine-tile image selection and collage-based object localization, using 262 images from 10 recognizable classes. Following [21], we evaluate CLIP ViT-B/32 on grids and OWLv2 on collages,

Table 5: Transferability of SMER and SMER+DES-Adv to multimodal foundation models ($N = 1000$). ASR is computed over samples correctly recognized in their benign form.

Model	Attack	Top-1			Top-5		
		Ben.	Adv.	ASR	Ben.	Adv.	ASR
Claude 4.6	SMER	47.20	25.70	48.30	70.00	41.70	42.14
	SMER+DES-Adv	47.20	**20.80**	**57.62**	70.00	**40.00**	**44.28**
GPT-5.6	SMER	42.90	17.80	60.84	67.70	32.90	53.91
	SMER+DES-Adv	42.90	**11.60**	**75.76**	67.70	**24.90**	**65.58**
Gemini 3.6 Flash	SMER	56.90	45.30	26.70	84.50	70.70	19.40
	SMER+DES-Adv	56.90	**37.20**	**40.24**	84.50	**61.90**	**29.11**

Table 6: CAPTCHA solving rate (CSR, %) across 500 sessions per setting; lower is better.

CAPTCHA Solver	Layout	Benign	Adversarial CSR	
		CSR	$\epsilon = 8/255$	$\epsilon = 16/255$
CLIP ViT-B/32	Grid	92.20	56.00	28.80
OWLv2	Collage	70.20	56.20	42.40
YOLOv8	Grid	92.40	49.80	30.60
	Collage	38.20	28.60	14.20
Qwen2.5-VL-3B-Instruct	Grid	99.00	73.60	46.40
	Collage	44.00	32.80	20.00
LLaVA-1.5-7B	Grid	99.40	74.00	43.40
	Collage	35.20	31.60	22.00
Halligan (GPT-4o)	Grid	94.00	67.00	37.40
	Collage	16.43	10.44	11.69

alongside YOLOv8, Qwen2.5-VL, LLaVA [17], and a GPT-4o-based Halligan solver [23]. Across 500 sessions per setting, we measure CAPTCHA solving rate (CSR) under benign and adversarial conditions ($\epsilon = 8/255, 16/255$). At $\epsilon = 16/255$, CSR drops from 92.2% to 28.8% for CLIP and from 70.2% to 42.4% for OWLv2 (Table 6). Figure 6 illustrates both formats. These experiments concern custom challenges, not production CAPTCHA services.

5.4 Ablation Study

Selection-Signal Ablation. We ablate local competence (C), gradient diversity (G), and noise sensitivity (S) on ImageNet using EnsAttack, six candidate surrogates (four selected), and nine unseen targets. Fixed uses Set 1; Random selects four models per

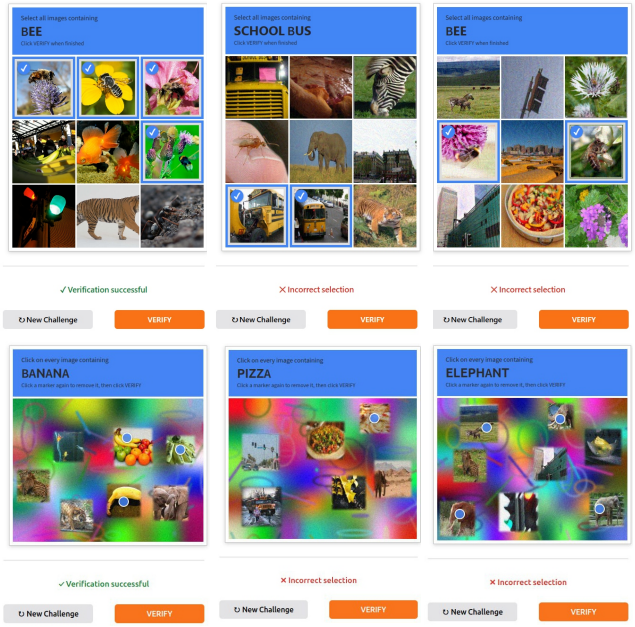


Figure 6: Examples of custom visual CAPTCHA challenges. Top: grid-based image selection; bottom: collage-based object localization. Columns show (a) benign challenges, (b) adversarial challenges at $\epsilon = 8/255$, and (c) adversarial challenges at $\epsilon = 16/255$.

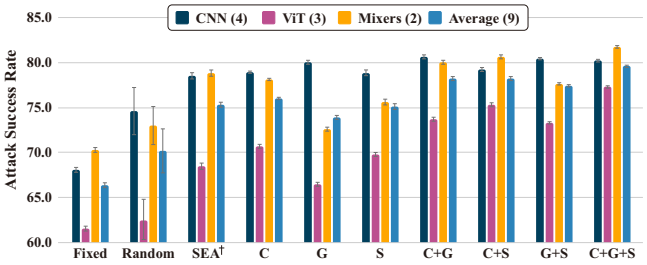


Figure 7: Ablation on ImageNet using EnsAttack (ASR, %). Bars show mean ASR, with error bars indicating standard deviation across five runs. Fixed denotes Set 1; Mixers include MLP-Mixer-B and ConvMixer. SEA† is independently reimplemented using per-iteration surrogate re-sampling without author-released code.

image; and SEA [29] re-samples four per iteration using our independent approximation. All variants exclude optional refinements and are evaluated over five runs.

As shown in Figure 7, the full C+G+S selector achieves 79.63% average ASR, outperforming Fixed, Random, and SEA† by 13.25, 9.41, and 4.34 percentage points, respectively. Removing C, G, or S reduces ASR to 77.42%, 78.28%, and 78.21%, respectively. The largest gain over SEA occurs on ViT targets (8.80 pp). The relatively small standard deviations indicate consistent performance across runs, supporting the complementary contributions of the three selection signals.

Table 7: Online DES-Adv selection time for all 1,000 ImageNet images (seconds).

Stage	Time	Stage	Time
Query embedding	7.505	Noisy gradients (S)	83.874
FAISS retrieval	5.054	Gradient cosines	0.346
Competence (C)	0.131	Selection ranking	0.367
Clean gradients (G)	83.541	**Total**	**180.818**

Computational Overhead. As shown in Table 7, DES-Adv requires only 180.82 seconds (3.01 minutes) to select surrogate ensembles for all 1,000 ImageNet images combined (0.18 seconds per image). In contrast, MI-FGSM-based SMER attack generation takes 464.3, 546.3, and 670.0 minutes with fixed Sets 1–3, respectively, compared with 682.0 minutes using DES-Adv-selected surrogates (Table 10). The 12-minute difference between Set 3 and DES-Adv in attack generation is consistent with the varying computational costs of selected surrogate architectures, rather than additional selection overhead. Moreover, precomputed DSEL embeddings and surrogate logits enable efficient retrieval and competence estimation. This one-time offline preparation requires only 7.71 minutes and is reported separately in Table 13.

5.5 Scope and Limitations

Our evaluation uses images correctly classified by all targets, which may not reflect real Web traffic. Target predictions are used only for offline test-set selection, preserving query-free attack generation. DES-Adv requires ground-truth labels and a labeled DSEL, limiting label-free applicability. Web API results depend on lexical label matching and sometimes small conditional sample sizes. Dynamic selection incurs additional computation, while service-side resizing and compression remain unexplored. CAPTCHA experiments use custom challenges rather than production services. Future work should address realistic deployments and matched computational budgets.

6 Conclusion

We investigated whether fixed surrogate composition limits transferable adversarial attacks against unseen Web vision models. We introduced DES-Adv, an input-adaptive framework that selects image-specific surrogate subsets using local competence, gradient diversity, and noise sensitivity, without modifying the underlying ensemble attack. Experiments on ImageNet, CIFAR-10, and CIFAR-100 demonstrate improved transferability across ensemble attack methods and heterogeneous target architectures, outperforming the evaluated fixed surrogate combinations.

Evaluations on commercial vision APIs, hosted models, and multimodal interfaces further demonstrate the relevance of adaptive surrogate selection to Web-facing systems. Separate experiments examine the effects of adversarial imagery on automated solvers for custom CAPTCHA challenges. Our findings highlight surrogate composition as an important variable in black-box robustness evaluation rather than a fixed design choice. Future work will investigate label-free selection, deployment-induced preprocessing shifts, and improved computational efficiency.

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A Surrogate Model Combinations.

Tables 8 and 9 list all 15 possible combinations of selecting 4 surrogate models from a pool of 6 for ImageNet and CIFAR benchmarks, respectively. While the main paper reports results on a subset of representative combinations, here we include all combinations for completeness.

B Additional Experimental Results

This appendix provides supporting results for the main experiments, including predefined surrogate combinations, detailed target-wise comparisons on ImageNet and CIFAR-10/100, and additional analyses of the optional transfer-failure refinements.

B.1 Detailed Results for Figure 2

Table 10 provides a detailed version of Figure 2. It reports attack success rates for all nine individual target models grouped into three architecture families: CNN, ViT, and MLP.

The table includes results for different ensemble attack methods (Ens. Attack, SVRE, AdaEA, and SMER) evaluated across three predefined surrogate sets. For each configuration, we report the mean and standard deviation over five runs. In addition to per-model results, the table also summarizes the average ASR, along with noise rate, pixel difference, and computation time.

For DES-Adv, the table reports the average ASR together with the relative gain compared to the best-performing surrogate set among the three predefined sets.

B.2 Offline DSEL Precomputation

DES-Adv precomputes DINO embeddings and surrogate logits for 30,000 ImageNet DSEL images. As shown in Table 13, this one-time

Table 8: Surrogate model combinations used for each set on ImageNet.

Set	Model 1	Model 2	Model 3	Model 4
Set 1	ResNet-18	Inception-V3	DeiT-Tiny	ViT-Tiny
Set 2	Inception-V3	DeiT-Tiny	ViT-Tiny	EfficientNet-B0
Set 3	Inception-V3	DeiT-Tiny	XCiT-Tiny	EfficientNet-B0
Set 4	ResNet-18	Inception-V3	ViT-Tiny	XCiT-Tiny
Set 5	ResNet-18	Inception-V3	ViT-Tiny	EfficientNet-B0
Set 6	ResNet-18	Inception-V3	XCiT-Tiny	EfficientNet-B0
Set 7	ResNet-18	DeiT-Tiny	ViT-Tiny	XCiT-Tiny
Set 8	ResNet-18	DeiT-Tiny	ViT-Tiny	EfficientNet-B0
Set 9	ResNet-18	DeiT-Tiny	XCiT-Tiny	EfficientNet-B0
Set 10	ResNet-18	ViT-Tiny	XCiT-Tiny	EfficientNet-B0
Set 11	Inception-V3	DeiT-Tiny	ViT-Tiny	XCiT-Tiny
Set 12	ResNet-18	Inception-V3	DeiT-Tiny	XCiT-Tiny
Set 13	ResNet-18	Inception-V3	DeiT-Tiny	EfficientNet-B0
Set 14	Inception-V3	ViT-Tiny	XCiT-Tiny	EfficientNet-B0
Set 15	DeiT-Tiny	ViT-Tiny	XCiT-Tiny	EfficientNet-B0

Table 9: Surrogate model combinations used for each set on CIFAR-10/100.

Set	Model 1	Model 2	Model 3	Model 4
Set 1	ResNet-18	Inception-V3	DeiT-Tiny	ViT-Tiny
Set 2	Inception-V3	DeiT-Tiny	ViT-Tiny	EfficientNet-B0
Set 3	Inception-V3	DeiT-Tiny	GCViT-Tiny	EfficientNet-B0
Set 4	ResNet-18	Inception-V3	ViT-Tiny	GCViT-Tiny
Set 5	ResNet-18	Inception-V3	ViT-Tiny	EfficientNet-B0
Set 6	ResNet-18	Inception-V3	GCViT-Tiny	EfficientNet-B0
Set 7	ResNet-18	DeiT-Tiny	ViT-Tiny	GCViT-Tiny
Set 8	ResNet-18	DeiT-Tiny	ViT-Tiny	EfficientNet-B0
Set 9	ResNet-18	DeiT-Tiny	GCViT-Tiny	EfficientNet-B0
Set 10	ResNet-18	ViT-Tiny	GCViT-Tiny	EfficientNet-B0
Set 11	Inception-V3	DeiT-Tiny	ViT-Tiny	GCViT-Tiny
Set 12	ResNet-18	Inception-V3	DeiT-Tiny	GCViT-Tiny
Set 13	ResNet-18	Inception-V3	DeiT-Tiny	EfficientNet-B0
Set 14	Inception-V3	ViT-Tiny	GCViT-Tiny	EfficientNet-B0
Set 15	DeiT-Tiny	ViT-Tiny	GCViT-Tiny	EfficientNet-B0

Table 10: Detailed comparison of ensemble attacks with different sets and DES-Adv for Figure 2 on ImageNet.

Attack	Method	Set	CNN				ViT			Other		Avg ASR	Noise Rate	Pixel-diff	Time
			Res-152	WRN101-2	RegNet-320	VGG-19	ViT-B	DeiT-B	Swin-B	MLP-Mixer-B	ConvMixer				
MI-FGSM	Ens. Attack	Set 1	58.70±0.10	75.98±0.16	51.64±0.29	85.86±0.09	63.20±0.01	79.78±0.04	41.18±0.18	73.94±0.09	70.00±0.12	66.70±0.05	0.190 ± 0.080	0.025 ± 0.001	18.6±0.6
	SVRE	Set 1	56.08±0.10	74.64±0.16	49.18±0.29	86.72±0.09	59.12±0.01	77.36±0.04	39.52±0.18	70.94±0.09	67.52±0.12	64.56±0.05	0.190 ± 0.080	0.025 ± 0.001	25.4±0.6
	AdaEA	Set 1	59.36±0.84	77.90±0.80	51.78±0.97	87.66±1.04	69.10±1.18	85.56±0.79	45.78±0.74	77.52±1.28	70.02±0.89	69.41±0.30	0.180 ± 0.077	0.024 ± 0.001	58.4±0.6
	SMER	Set 1	71.47±1.53	85.50±0.35	64.30±1.01	92.73±0.42	77.70±0.70	92.57±0.25	54.23±0.31	86.60±0.17	80.13±0.96	78.36±0.07	0.197 ± 0.080	0.027 ± 0.001	464.3±18.3
	Ens. Attack	Set 2	60.02±0.25	76.74±0.23	59.14±0.15	86.80±0.14	68.14±0.13	84.08±0.19	50.86±0.13	77.92±0.08	70.82±0.16	**70.5±0.06**	0.190 ± 0.080	0.026 ± 0.001	21.4±1.7
	SVRE	Set 2	57.96±0.76	75.74±0.80	55.46±0.63	86.82±0.76	64.90±1.53	81.4±0.57	48.20±0.80	75.90±0.66	70.24±1.04	68.51±0.27	0.190 ± 0.080	0.025 ± 0.001	31.2±1.3
	AdaEA	Set 2	60.12±0.63	79.88±0.36	59.70±0.57	88.78±0.78	72.66±0.83	88.40±0.31	54.14±1.05	81.58±1.00	73.20±0.37	**73.16±0.39**	0.177 ± 0.076	0.024 ± 0.001	71.4±1.8
	SMER	Set 2	71.60±1.15	86.73±0.81	70.83±0.40	93.10±0.53	80.80±1.32	93.33±0.35	61.23±0.70	88.80±0.26	82.50±0.66	80.99±0.46	0.197 ± 0.080	0.027 ± 0.001	546.3±24.9
	Ens. Attack	Set 3	65.08±0.27	78.94±0.21	66.66±0.15	87.78±0.11	58.86±0.11	70.54±0.11	54.32±0.13	69.64±0.15	74.64±0.11	69.61±0.04	0.192 ± 0.081	0.026 ± 0.001	28.6±0.6
	SVRE	Set 3	63.12±0.40	77.72±0.64	66.64±1.09	87.94±0.48	56.68±0.34	69.82±1.29	53.08±0.94	68.42±0.71	73.60±0.56	**68.56±0.16**	0.191 ± 0.081	0.025 ± 0.001	39.8±2.2
	AdaEA	Set 3	66.42±0.33	80.44±0.96	71.18±0.68	90.08±0.29	62.54±1.13	74.50±0.42	58.60±0.65	73.28±0.59	76.92±0.59	72.66±0.35	0.180 ± 0.077	0.024 ± 0.001	94.4±4.2
	SMER	Set 3	78.23±0.59	88.23±0.9	81.68±0.56	94.48±0.51	72.65±0.61	84.83±0.61	67.95±0.51	82.45±0.44	85.43±0.92	**81.77±0.17**	0.198 ± 0.081	0.027 ± 0.001	670.0±3.9
PGD	Ens. Attack	Set 1	42.94±0.36	66.04±0.3	38.28±0.13	78.56±0.21	43.88±0.18	65.6±0.29	29.72±0.2	59.78±0.16	53.22±0.08	53.11±0.07	0.119 ± 0.055	0.017 ± 0.001	18.4±0.6
	SVRE	Set 1	40.82±1.28	61.76±1.19	34.24±0.37	77.56±0.23	41.26±0.32	60.94±0.58	25.78±0.29	55.22±0.28	51.54±1.09	49.9±0.15	0.123 ± 0.059	0.017 ± 0.001	26.6±1.5
	AdaEA	Set 1	46.54±0.67	68.5±0.93	38.62±0.76	81.38±0.63	49.78±0.66	71.16±0.86	32.42±0.56	63.18±0.53	57.60±1.13	56.58±0.25	0.118 ± 0.056	0.017 ± 0.001	63.6±3.8
	SMER	Set 1	65.17±0.47	82.7±0.35	58.77±0.92	91.53±0.95	71.97±0.75	90.30±0.30	48.10±0.53	82.47±0.51	75.27±0.38	74.03±0.12	0.132 ± 0.055	0.019 ± 0.001	451.7±29.4
	Ens. Attack	Set 2	45.22±0.18	66.46±0.17	45.38±0.34	78.82±0.20	50.12±0.26	70.74±0.11	38.16±0.34	64.78±0.29	57.54±0.17	**57.47±0.05**	0.118 ± 0.011	0.017 ± 0.001	22.2±0.8
	SVRE	Set 2	41.3±0.47	62.48±0.54	40.52±0.43	77.00±1.63	47.38±0.36	66.74±0.66	34.40±0.85	62.36±0.87	56.20±0.50	54.26±0.38	0.124 ± 0.059	0.017 ± 0.001	31.8±1.6
	AdaEA	Set 2	46.70±0.87	68.56±0.62	45.64±0.59	81.92±0.70	57.74±0.77	79.00±0.71	41.86±0.61	71.72±0.52	61.36±0.82	**61.61±0.23**	0.118 ± 0.055	0.018 ± 0.001	71.8±4.5
	SMER	Set 2	64.50±1.30	82.90±0.70	64.10±0.44	92.37±0.68	75.10±0.56	91.6±0.35	55.73±0.93	85.17±0.98	77.27±0.93	76.53±0.37	0.132 ± 0.054	0.020 ± 0.001	566.0±46.6
	Ens. Attack	Set 3	48.52±0.15	66.48±0.13	53.90±0.10	80.54±0.19	41.48±0.23	53.72±0.15	41.80±0.23	57.40±0.17	60.94±0.13	56.09±0.06	0.118 ± 0.056	0.016 ± 0.001	27.8±1.6
	SVRE	Set 3	46.56±0.59	64.08±0.58	51.40±1.69	79.34±0.59	39.34±0.83	52.92±0.48	38.22±0.61	54.54±0.74	59.36±0.44	**53.97±0.15**	0.124 ± 0.060	0.017 ± 0.001	38.6±1.1
	AdaEA	Set 3	51.68±0.96	69.82±0.5	58.10±0.58	83.88±0.15	45.36±0.52	60.30±0.46	45.54±0.90	60.66±0.75	65.30±0.53	60.07±0.11	0.118 ± 0.056	0.017 ± 0.001	92.2±2.4
	SMER	Set 3	72.57±0.35	85.43±0.29	77.40±0.79	93.60±0.10	67.13±0.96	80.43±0.15	63.67±0.57	77.63±0.38	81.97±0.61	**77.76±0.11**	0.134 ± 0.056	0.019 ± 0.001	689.0±58.4
MI-FGSM	Ens. Attack	DES-Adv	69.75±0.06	82.90±0.08	76.68±0.05	92.00±0.08	76.70±0.22	88.60±0.14	66.70±0.22	83.10±0.14	80.48±0.05	**79.66 (+9.16 ↑)**	0.199 ± 0.081	0.026 ± 0.001	34.4±1.7
	SVRE	DES-Adv	69.35±0.91	82.08±0.62	75.70±0.77	92.38±0.61	75.35±0.57	88.8±0.37	65.63±0.61	83.48±0.68	80.80±0.64	**79.28 (+10.72 ↑)**	0.201 ± 0.081	0.026 ± 0.001	45.4±0.9
	AdaEA	DES-Adv	69.17±0.76	83.23±0.64	75.07±1.11	93.13±0.25	77.03±0.86	90.1±0.62	66.57±0.81	84.77±0.29	82.47±0.68	**80.17 (+7.01 ↑)**	0.189 ± 0.077	0.025 ± 0.001	98.0±4.8
	SMER	DES-Adv	79.43±0.25	88.97±0.98	83.81±0.99	95.48±0.38	83.63±0.50	93.97±0.67	75.07±0.52	88.93±0.21	88.78±0.47	**86.45 (+4.68 ↑)**	0.206 ± 0.080	0.029 ± 0.001	682.0±13.0
PGD	Ens. Attack	DES-Adv	61.78±0.05	78.95±0.34	69.93±0.21	90.25±0.06	68.03±0.22	83.98±0.15	59.15±0.33	77.98±0.17	74.83±0.17	**73.87 (+16.40 ↑)**	0.129 ± 0.056	0.018 ± 0.001	33.2±2.4
	SVRE	DES-Adv	58.50±0.36	75.70±0.56	66.13±1.07	89.03±0.21	65.20±0.78	81.83±0.59	57.63±0.47	76.80±0.36	73.70±0.69	**71.61 (+17.64 ↑)**	0.133 ± 0.058	0.018 ± 0.001	41.6±1.7
	AdaEA	DES-Adv	61.43±0.50	77.97±0.45	69.73±0.76	91.10±0.26	68.17±0.76	84.80±0.17	60.37±1.07	78.93±0.60	74.87±0.38	**74.15 (+12.54 ↑)**	0.128 ± 0.056	0.017 ± 0.001	94.4±1.9
	SMER	DES-Adv	79.01±0.88	89.78±0.80	84.23±0.35	96.15±0.12	82.87±0.40	94.19±0.31	72.92±0.78	90.41±0.29	88.28±0.76	**86.43 (+8.67 ↑)**	0.143 ± 0.056	0.020 ± 0.002	721.3±8.1

preparation requires 462.80 seconds (7.71 minutes). The cached representations and predictions are reused for local competence estimation without repeated DSEL inference. This cost is excluded from online selection and attack-generation times.

C Web-Service Evaluation Protocol

C.1 Conditional ASR and Sample Sizes

For each image i, let $b_{i,k}$ and $a_{i,k}$ indicate whether its reference class matches any benign or adversarial Top-k prediction, respectively. We compute conditional attack success as

$$\mathrm{ASR@}k = \frac{\sum_{i=1}^{N} b_{i,k}(1 - a_{i,k})}{\sum_{i=1}^{N} b_{i,k}}. \tag{12}$$

The denominator contains only benign-matched inputs, and is identical for both compared attacks. The Ben. and Adv. columns instead report agreement over all $N = 1000$ inputs; their difference therefore does not directly determine conditional ASR.

C.2 Label Matching and Output Interpretation

Unlike ImageNet classifiers, Web vision APIs return free-form labels that vary in specificity and vocabulary. We compare each returned

Table 11: Comparison of ensemble attack methods across different surrogate sets and DES-Adv on CIFAR-10 using the MI-FGSM.

Method	Pool	CNN			ViT			Conv.	Avg ASR	Noise Rate	Pixel-diff	Time (m)
		BiT-101	Res-152	RegNet-160	ViT-B	DeiT-B	Swin-B	ConvMixer				
Ens. Attack	Set 1	94.60±0.20	91.57±0.75	87.17±1.07	66.33±0.58	73.17±1.26	80.37±0.55	91.37±0.21	**83.51±0.34**	0.093 ± 0.058	0.025 ± 0.001	18.3±2.5
SVRE	Set 1	93.2±0.46	88.87±0.55	84.17±1.29	69.4±1.73	75.17±0.76	76.5±0.5	88.97±0.55	**82.32±0.42**	0.095 ± 0.060	0.025 ± 0.001	24.7±2.1
AdaEA	Set 1	92.93±0.68	90.70±0.10	86.00±1.21	66.97±1.55	76.57±0.29	79.20±0.26	90.87±0.93	**83.32±0.26**	0.087 ± 0.056	0.024 ± 0.002	59.3±1.5
SMER	Set 1	83.10±0.20	81.53±0.38	76.93±0.59	68.80±1.18	74.40±0.82	76.60±0.36	80.67±0.38	**77.43±0.41**	0.096 ± 0.071	0.025 ± 0.001	406.7±4.9
Ens. Attack	Set 2	93.60±0.20	86.57±0.15	87.17±0.38	65.93±0.35	74.63±0.25	78.20±0.26	88.43±0.15	82.08±0.11	0.092 ± 0.057	0.025 ± 0.001	21.3±1.5
SVRE	Set 2	90.57±0.51	83.83±0.42	82.20±0.79	63.50±0.78	71.63±0.35	73.57±0.51	84.00±0.30	78.47±0.10	0.095 ± 0.060	0.025 ± 0.001	27.3±2.5
AdaEA	Set 2	91.73±0.21	86.23±0.42	83.57±0.31	63.57±0.40	71.73±0.31	75.30±0.26	86.77±0.15	79.84±0.11	0.086 ± 0.056	0.023 ± 0.003	67.7±2.1
SMER	Set 2	79.60±0.62	77.17±0.45	76.03±0.21	66.30±0.56	71.43±0.40	72.20±0.26	76.87±0.68	74.23±0.18	0.094 ± 0.073	0.024 ± 0.001	416.3±8.1
Ens. Attack	Set 3	94.70±0.17	87.87±0.06	87.50±0.17	60.43±0.50	68.63±0.21	78.70±0.30	89.7±0.10	81.08±0.09	0.091 ± 0.057	0.025 ± 0.001	24.3±2.1
SVRE	Set 3	92.23±0.81	86.67±0.38	84.27±0.64	58.70±0.79	64.57±0.38	75.00±0.20	87.27±0.40	78.39±0.20	0.094 ± 0.060	0.025 ± 0.001	31.3±2.1
AdaEA	Set 3	93.27±0.15	87.50±0.10	85.93±0.42	59.10±1.01	66.67±0.50	78.03±0.15	89.63±0.06	80.02±0.14	0.086 ± 0.055	0.023 ± 0.003	77.3±2.0
SMER	Set 3	84.87±0.06	81.97±0.06	79.77±0.46	63.93±1.11	69.67±0.46	77.00±0.10	81.93±0.58	77.02±0.26	0.097 ± 0.071	0.025 ± 0.009	478.3±9.6
Ens. Attack	DES-Adv	97.43±0.06	96.47±0.06	93.60±0.10	78.47±0.15	85.20±0.10	88.67±0.15	95.73±0.06	**90.80 (+7.29 ↑)**	0.102 ± 0.057	0.027 ± 0.001	26.2±1.7
SVRE	DES-Adv	96.10±0.26	94.43±0.15	90.93±0.75	75.47±0.25	81.03±0.29	86.23±0.75	93.27±0.38	**88.21 (+5.89 ↑)**	0.101 ± 0.060	0.026 ± 0.001	33.6±2.0
AdaEA	DES-Adv	96.10±0.26	95.53±0.15	92.43±0.64	77.43±0.97	82.53±0.38	87.77±0.21	95.03±0.58	**89.55 (+6.23 ↑)**	0.094 ± 0.056	0.024 ± 0.002	81.2±2.5
SMER	DES-Adv	94.60±0.50	91.23±0.25	85.37±0.86	85.03±0.50	89.57±0.31	91.07±0.40	91.43±0.31	**89.76 (+12.33 ↑)**	0.109 ± 0.069	0.028 ± 0.007	499.1±12.8

Table 12: Comparison of ensemble attack methods across different surrogate sets and DES-Adv on CIFAR-100 using MI-FGSM.

Attack	Pool	CNN			ViT			MLP	Avg ASR	Noise Rate	Pixel-diff	Time (m)
		BiT-101	Res-152	RegNet-320	ViT-B	DeiT-B	Swin-B	ConvMixer				
Ens. Attack	Set 1	79.70±0.10	71.23±0.12	86.33±0.06	71.15±0.05	81.10±0.10	78.47±0.06	88.60±0.10	79.51±0.04	0.087+0.052	0.024+0.001	19.0±1.7
SVRE	Set 1	78.03±0.40	68.90±1.14	83.33±0.81	70.57±0.15	78.30±1.13	75.50±0.26	85.1±0.89	77.10±0.50	0.089+0.054	0.025+0.001	24.3±0.6
AdaEA	Set 1	80.90±0.50	71.93±0.58	86.00±0.62	70.20±0.36	77.27±0.31	76.30±0.30	87.77±0.45	78.62±0.17	0.082+0.051	0.023+0.003	64.7±0.6
SMER	Set 1	82.73±0.59	76.77±0.9	87.77±0.76	76.70±1.23	80.17±1.01	80.07±0.76	89.40±0.50	81.94±0.73	0.103+0.060	0.028+0.002	472.0±5.0
Ens. Attack	Set 2	78.83±0.06	72.70±0.01	85.08±0.03	71.00±0.10	82.22±0.03	81.17±0.06	87.20±0.10	79.74±0.01	0.087+0.052	0.024+0.001	21.3±1.5
SVRE	Set 2	77.07±1.16	70.60±0.90	82.13±0.80	70.97±0.06	79.63±0.65	78.33±0.49	85.83±0.68	77.80±0.25	0.089+0.054	0.025+0.001	32.3±0.6
AdaEA	Set 2	79.33±0.42	72.5±0.56	85.47±0.23	70.63±0.51	79.80±0.56	78.97±0.25	87.97±0.61	79.24±0.09	0.083+0.051	0.023+0.001	75.7±1.5
SMER	Set 2	82.00±0.61	76.87±0.57	86.43±0.38	77.17±0.32	82.23±0.47	80.3±0.52	89.77±0.29	82.11±0.26	0.103+0.060	0.028+0.002	551.0±10.0
Ens. Attack	Set 3	80.23±0.12	76.00±0.01	86.43±0.12	69.67±0.06	83.03±0.06	82.20±0.10	90.12±0.03	**81.10±0.03**	0.087+0.052	0.024+0.001	32.7±1.5
SVRE	Set 3	80.37±0.55	76.73±1.51	84.70±1.21	69.43±0.67	82.53±1.19	81.30±0.61	88.97±1.76	**80.58±0.59**	0.090+0.054	0.025+0.001	45.7±0.6
AdaEA	Set 3	82.03±1.36	77.63±1.36	87.10±0.95	70.27±0.67	83.27±0.85	82.97±0.06	90.93±0.35	**82.03±0.76**	0.083+0.051	0.023+0.003	109.5±3.5
SMER	Set 3	85.37±1.20	83.90±0.98	89.40±0.10	77.40±0.36	86.43±3.54	84.27±0.38	92.83±2.04	**85.66±1.05**	0.103+0.059	0.028+0.002	672.3±12.5
Ens. Attack	DES-Adv	88.47±0.15	85.33±0.06	93.17±0.15	77.60±0.20	88.93±0.15	90.40±0.10	94.00±0.35	**88.27 (+7.17 ↑)**	0.095+0.052	0.025+0.001	23.0±1.0
SVRE	DES-Adv	87.37±0.42	83.07±0.35	90.87±0.32	76.37±0.12	86.33±0.25	87.90±0.30	92.33±0.64	**86.32 (+5.74 ↑)**	0.098+0.054	0.027+0.001	48.3±5.5
AdaEA	DES-Adv	88.47±0.55	85.03±0.25	91.93±0.67	77.13±0.35	87.83±0.21	88.97±0.25	93.17±0.25	**87.50 (+5.47 ↑)**	0.091+0.051	0.024+0.001	118.7±5.0
SMER	DES-Adv	91.67±0.38	87.00±0.61	93.23±0.15	82.83±0.40	90.50±0.62	91.90±1.01	95.07±0.60	**90.31 (+4.65 ↑)**	0.111+0.059	0.029+0.002	702.0±31.2

Table 13: One-time offline precomputation for 30,000 ImageNet DSEL images (seconds).

Stage	Time	Stage	Time
DSEL embeddings	156.95	XCiT-Tiny logits	95.40
ResNet-18 logits	41.76	Inception-V3 logits	43.22
DeiT-Tiny logits	41.69	ViT-Tiny logits	41.81
EfficientNet-B0 logits	41.97	–	–
		Total offline time	**462.80**

Table 14: Benign-matched sample counts for conditional ASR ($N = 1000$).

Service	Top-1	Top-5	Top-10	Service	Top-1	Top-5	Top-10
AWS	89	476	639	Claude 4.6	472	700	–
Google CV	117	395	554	GPT-5.6	429	677	–
HF ConvNeXt	984	1000	1000	Gemini 3.6	569	845	–
Imagga	509	652	708				

label with the ImageNet reference class using normalized lexical matching, including shared-word and substring matches. For example, "tabby" can match "tabby cat," whereas a generic label such as "bird" need not match the fine-grained class "goldfinch." The same matching rules are applied to benign and adversarial predictions, with Top-k agreement requiring at least one match among the first k labels.

Conditional ASR is computed only over benign-matched inputs, as defined in Eq. 12. Consequently, effective sample sizes differ across services and rank cutoffs (Table 14). These measurements reflect agreement under our lexical matching protocol rather than each service's native classification accuracy. Multimodal models are prompted to return five ranked categories, evaluated using the same reference taxonomy.

D Transfer-Failure Analysis

Observed Transfer-Failure Patterns. We examine two recurring patterns of limited adversarial transferability. First, complex-background images may exhibit weak perturbations that fail to transfer reliably to unseen targets (Figure 8(a)). Second, simple-background images may fool surrogate models while remaining correctly classified by unseen targets (Figure 8(b)). These observations motivate the optional refinements evaluated below, without establishing background complexity as a causal factor.

Table 15: Effect of optional perturbation amplification on transfer ASR for ImageNet Failure Case 1. Increased perturbation budgets should not be attributed to surrogate selection alone.

Method	CNN Models				ViT Models			Other Targets		Avg ASR	Noise Rate	Pixel-diff	Time
	Res152	WRN101-2	RegNet320	VGG19	ViT-B	DeiT-B	Swin-B	Mixer-B	ConvMixer				
Ens. Attack	32.80	51.00	33.40	58.60	51.60	69.20	34.80	54.40	44.00	47.76	0.064±0.013	0.026±0.001	3
SVRE	31.60	50.40	30.40	59.20	49.40	65.40	35.40	54.00	43.00	46.53	0.063±0.012	0.026±0.001	6
AdaEA	34.40	51.40	33.00	64.80	55.80	72.40	38.40	60.40	44.80	50.60	0.059±0.012	0.024±0.001	7
SMER	38.40	61.80	39.40	73.00	64.00	80.40	44.80	68.20	53.40	58.16	0.070±0.013	0.028±0.001	40
Ens. Attack + DES-Adv	54.40	71.00	54.60	74.80	78.60	86.40	58.20	77.20	66.60	69.09	0.122±0.006	0.039±0.005	3
SVRE + DES-Adv	52.60	68.60	53.40	77.40	77.00	84.60	58.20	76.80	69.00	68.62	0.122±0.007	0.039±0.005	6
AdaEA + DES-Adv	58.80	75.40	61.20	82.40	81.40	90.80	65.40	82.40	71.00	74.31	0.122±0.008	0.038±0.005	7
SMER + DES-Adv	66.00	79.00	64.20	86.60	83.80	92.80	68.60	86.60	76.20	78.20	0.122±0.007	0.039±0.005	40
Ens. Attack + DES-Adv^2	67.80	77.00	70.20	83.80	86.40	90.60	71.60	84.20	75.40	78.56	0.180±0.009	0.050±0.006	2
SVRE + DES-Adv^2	67.20	75.60	68.80	83.40	83.60	89.80	68.00	84.20	75.60	77.36	0.180±0.009	0.050±0.007	6
AdaEA + DES-Adv^2	74.20	83.80	74.40	89.00	90.00	94.60	77.80	90.20	80.80	83.87	0.180±0.011	0.049±0.007	7
SMER + DES-Adv^2	78.60	85.80	79.20	93.20	92.80	95.60	79.80	92.40	83.80	86.80	0.181±0.009	0.051±0.007	40

Table 16: Effect of optional surrogate-ensemble expansion on transfer ASR for ImageNet Failure Case 2. The enlarged ensemble changes the computational cost and active ensemble size.

Attack	CNN Models				ViT Models			Other Targets		Avg ASR	Noise Rate	Pixel-diff	Time
	Res-152	WRN101-2	RegNet-320	VGG-19	ViT-B	DeiT-B	Swin-B	MLP-Mixer-B	ConvMixer				
Ens. Attack (4)	41.40	58.40	43.00	68.80	48.60	64.20	33.40	54.80	51.20	51.53	0.358 ± 0.056	0.025 ± 0.002	2
SVRE (4)	41.20	60.20	42.20	69.40	48.40	63.40	34.80	53.60	52.40	51.73	0.358 ± 0.055	0.025 ± 0.002	3
AdaEA (4)	40.60	57.80	43.80	71.40	52.40	70.80	39.00	59.60	52.60	54.22	0.333 ± 0.075	0.023 ± 0.003	6
SMER (4)	53.20	74.00	53.60	86.80	63.00	82.60	45.60	72.00	62.60	65.93	0.364 ± 0.054	0.026 ± 0.002	40
Ens. Attack (5)	56.00	68.40	58.20	83.80	56.00	71.60	47.20	59.40	63.80	62.71	0.363 ± 0.056	0.025 ± 0.002	2
SVRE (5)	52.40	65.20	59.00	83.20	52.00	70.40	46.60	58.60	61.20	60.96	0.364 ± 0.056	0.025 ± 0.002	3
AdaEA (5)	55.80	69.40	59.60	84.20	58.00	74.00	50.40	60.60	62.40	63.82	0.345 ± 0.056	0.023 ± 0.003	9
SMER (5)	72.40	81.60	76.00	94.60	72.20	88.00	63.00	78.00	76.80	78.07	0.369 ± 0.055	0.026 ± 0.002	63
Ens. Attack (6)	59.00	70.00	61.60	82.20	54.20	71.40	48.40	58.60	66.00	63.49	0.360 ± 0.056	0.025 ± 0.002	3
SVRE (6)	53.40	70.80	59.40	82.80	50.40	64.80	45.80	58.20	64.80	61.16	0.362 ± 0.056	0.025 ± 0.002	4
AdaEA (6)	60.80	74.20	63.20	86.00	56.20	74.80	50.60	64.20	68.40	66.49	0.336 ± 0.075	0.023 ± 0.004	13
SMER (6)	77.80	88.20	80.20	95.40	72.20	90.20	64.60	77.20	80.40	80.69	0.367 ± 0.054	0.026 ± 0.002	92



a) Failure Case 1 b) Failure Case 2

Figure 8: Representative transfer-failure patterns: (a) complex backgrounds with weak perturbations; (b) simple backgrounds where surrogate-successful attacks fail to transfer to unseen targets.

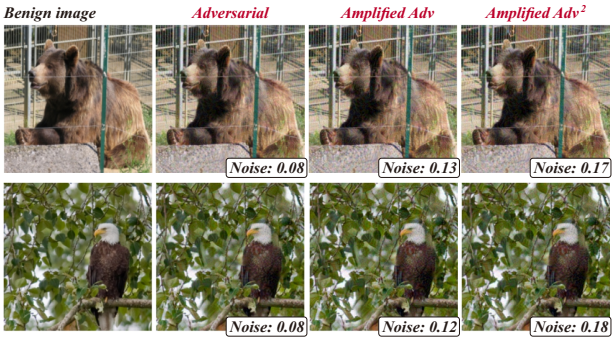


Figure 9: ImageNet examples before and after optional perturbation amplification. Noise rates increase with amplification; these examples are not matched-perturbation-budget comparisons.

First Failure Case. We evaluate 500 additional ImageNet images using four ensemble attacks with MI-FGSM (Set 2). The original attacks achieve 46.53–58.16% ASR, with noise rates of 0.059–0.070 (Table 15). To investigate insufficient perturbation strength, we separately evaluate doubling and tripling the generated perturbations. Average noise rates increase to 0.122 and 0.180, respectively, while ASR rises to 68.62–78.20% and 77.36–86.80%. These gains reflect increased perturbation magnitude and must not be interpreted as equal-budget improvements from surrogate selection alone.

Second Failure Case. We evaluate 500 additional ImageNet images with simple backgrounds, where surrogate-successful perturbations exhibit limited transferability. We investigate expanding the active ensemble from four to five or six surrogates. As shown in Table 16, average ASR increases from 51.53–65.93% with four models to 60.96–78.07% with five. Adding a sixth model yields smaller additional gains, reaching 61.16–80.69%, at greater computational cost. These results illustrate a transferability–cost trade-off rather than an equal-compute improvement.